SURFACE & PACKAGE STUDY

NEUROPIXELS 1.0 / REFERENCE ASSEMBLY AND PROCEDURAL CONTACT DETAIL

NPX–001

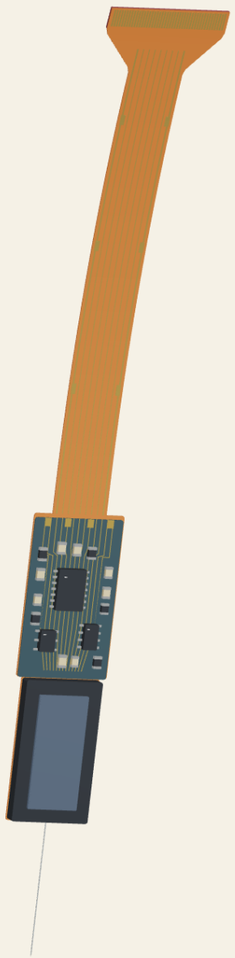
REV 01 / SHEET 02 OF 02

25 OCTOBER 2026

G Reference assembly

Published envelopes; reconstructed local geometry

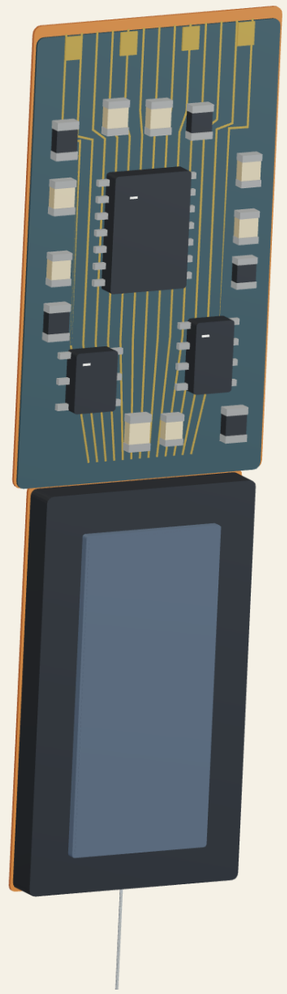
Flex shown at its 39.5 mm developed length.



10 mm

H Proximal package

Silicon cap, epoxy and populated flex base

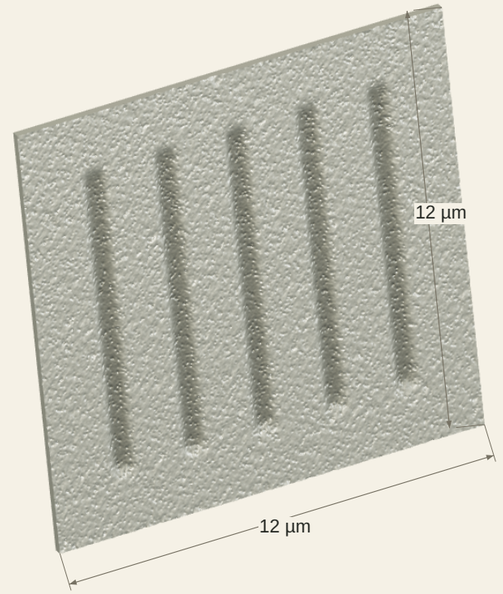


6.2 mm probe-base width / 7.2 mm SMD-base width.
3.9 × 8.5 mm silicon cap envelope.
Component and conductor layout is illustrative.

2 mm

I Porous TiN surface

High-resolution standalone contact model



2 µm

WHAT THE MODEL RESOLVES

Five SEM-inspired grooves are explicitly modeled.
Fine surface relief is procedural, with a fixed seed.
The detail contact contains 52,482 triangles.
An embedded normal map adds finer granularity.

Pore size, density, shape and layer thickness are not measured. This is not an electrochemical surface-area or impedance model.

MODEL VARIANTS

Drawing-matched study: reproduces the supplied illustration’s head outline and gold fanout.
Reference assembly: uses published package, flex and silicon-cap envelope dimensions.
Both share the same 10 mm shank, 960-site lattice and reconstructed tip reference electrode.
Gold-coloured conductors represent appearance, not a verified metallurgy stack or netlist.
The 45-contact flex terminal is inferred from the documented mating interface; pitch is assumed.

DELIVERABLES

GLB models in metres / editable builder in millimetres
STEP nominal shank and square contact footprints
Offline viewer / coordinates CSV / validation results
NOT FOR FABRICATION OR IMPLANTATION